

Radiance field-based 3D reconstruction and potential applications

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Abstract Radiance field-based 3D reconstruction has emerged as a transformative research direction due to its remarkable efficiency and quality. This paper presents a systematic analysis of representation models, reconstruction methodologies, and future applications in this field. We start from an overview of multi-view 3D reconstruction tasks, then focus on the key issue: how to represent 3D content effectively. Radiance fields are highlighted for their flexibility and representational completeness. Distinguished from the existing review literature, we adopt a multi-dimensional comparison between neural radiance fields (NeRF) and 3D Gaussian splatting (3DGS) to develop a unified and in-depth understanding of the radiance field-based approach. Beyond the initial goal of novel view synthesis (NVS), recent breakthroughs in geometry extraction are summarized. Finally, we explore potential applications across areas such as robot localization and mapping, virtual reality, physical simulation, and stereo display. Empowered by the flexible 3D representation within the radiance field-based paradigm, the latest advancements strive to push the boundaries and overcome long-standing bottlenecks in related domains.

Keywords 3D reconstruction, 3D representation, radiance fields, geometry extraction, 3D application

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1 Introduction

People have long sought the ability to record real-world scenes and reconstruct them accurately anytime and anywhere. The central challenge in this endeavor is known as three-dimensional (3D) reconstruction [1]. Humans naturally excel at perceiving and understanding their environment, primarily through the visual system [2]. Yet human memory is inherently limited in both capacity and duration. Traditional manual approaches to 3D reconstruction rely on the skilled manipulation of basic geometric elements such as points, lines, and planes, as well as their combinations [3]. In contrast, although machine sensing systems remain relatively primitive, they offer reliable storage and sharing of digital data.

Scanning-based solutions sample the structure or distribution of objects in the scene using specialized equipment,

including laser scanners [4,5], computed tomography [6,7], and structured light devices [8,9]. In contrast, image-based solutions [10,11] typically utilize multiple images captured from diverse viewpoints and positions within the scene. By employing computer vision (CV) algorithms [12], these techniques establish 3D correspondences across overlapping images to reconstruct both spatial structure and appearance information for each pixel. Due to their low equipment cost, data richness, and moderate computational requirements, image-based solutions have become a popular choice in 3D reconstruction.

Traditional 3D reconstruction can be traced back to the exploration of Binocular Depth Perception [13], which reconstructs depth by comparing two images of the same scene area, similar to human stereo vision. Binocular stereo reconstruction typically involves four key steps. (1) Feature extraction: Identifying local features (distinctive pixels) in each image, such as scale-invariant feature transform (SIFT) or ORB. (2) Feature matching: Establish correspondences

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based on the similarity of features. (3) Camera matrix estimation: Calculating camera poses (rotation and translation) from adjacent images captured by a continuously moving camera based on Epipolar geometry algorithms [14]. (4) Depth inference: Estimating depth by minimizing the reprojection error between adjacent images via triangulation. Structure from Motion (SfM) [15] extends binocular depth perception to multi-view scenarios captured by a continuously moving camera, employing an incremental optimization process. Bundle adjustment is commonly used to solve the SfM problem by minimizing an accumulated reprojection error. Multi-view stereo (MVS) [16] further extends the reconstruction from the feature level in SfM to the pixel level. For a reference image, related source images are selected based on sufficient overlap, and the depth of each pixel is calculated by optimizing the color similarity across image pairs. Ref. [16] also provides a complete and improved MVS pipeline, which is implemented as a widely used software named COLMAP. Traditional 3D reconstruction requires both local and global optimization based on pixel-wise correspondences in multiple images during the depth-inference process, resulting in complex algorithms.

The proven capabilities of machine learning in data prediction have naturally led to the development of learning-based methods for 3D reconstruction. These methods encode 3D spatial features within neural networks using multi-view images as input. The network parameters are optimized by comparing the 2D/3D outputs (such as images, depth maps, point clouds, or meshes) against the ground truth. An early and representative work is MVSNet [17], which constructs a 3D cost volume to store spatial features projected by differentiable homography. The loss is computed by comparing the predicted depth values of pixels with the ground truth. Subsequent improvements for MVSNet include: improving time and memory efficiency using the feature pyramid (Cascade-MVSNet [18], CVP-MVSNet [19]), using the patch match in MVS (PatchMatchNet [20]), or using the coarse-to-fine strategy (PointMVSNet [21]); improving the ability to extract features using a large model (MVSCRF [22], MVSNet++ [23], MVSFormer [24], TransMVSNet [25]). The review literature on MVSNet includes refs. [26–28]. MVSNet-based 3D reconstruction mainly aims at extracting features of depth derived from traditional MVS. As a supervised method [29], ground truth of 3D information is usually required, which, however, is not easily available in most cases. Recently, a completely different approach to 3D reconstruction has been proposed that encodes 3D spatial features using radiance fields [30,31] represented by neural radiance fields (NeRF) and 3D Gaussian splatting (3DGS). This approach is famous for the realism of rendering and the simplicity of the pipeline. During the training process, multi-view images with camera parameters are input. The output is predicted images with the same views as the input. Radiance

fields, as the 3D scene representation, are optimized through photometric loss between the corresponding input and output images without any additional 3D data. That is, what you see is what the 3D scene is. The differentiable nature of radiance field-based rendering ensures that loss can be back-propagated, enabling an efficient learning-based optimization. We will provide a more detailed introduction to this technical approach in Section 3.

The objective of this paper is to provide an understanding of the principles for newcomers and methodological references and inspiration for researchers. The rest of this paper is organized as follows: Section 2 discusses the common form of 3D representations. In Section 3, we introduce the basic architecture and related concepts for radiance field-based reconstruction from a comparative perspective. Section 4 summarizes the up-to-date work on 3D geometry reconstruction, which has been less addressed in the existing literature. In Section 5, we discuss potential directions and future applications empowered by the radiance field-based paradigm. Finally, a conclusion is drawn in Section 6. For simplicity, in subsequent discussions, we will also refer to the radiance field-based approach as modern 3D reconstruction, which takes multi-view images as its sole input to construct complete and continuous 3D scenes while enabling scene optimization through differentiable learning-based methods.

2 3D scene representation

3D scene representation, which decides how 3D spatial and appearance information is described and expressed, is a prerequisite for scene reconstruction. It can usually be classified into explicit and implicit based on visual accessibility, as shown in Figure 1.

Explicit representations include point clouds, voxels, and meshes. Point cloud represents surfaces in the scene with sampling points that can be recorded in a 3D coordinate system. A Cartesian coordinate point set can be described as $[(x_1, y_1, z_1), \dots, (x_n, y_n, z_n)]$, where n is the number of points. Each point is associated with an RGB color value if texture information is included. A point cloud is a common format for data output from 3D scanners. The advantage is that it is intuitive and simple, while the disadvantages include a large data volume and severe loss of local spatial information. The lack of point relationships also leads to the problem of ambiguity. A voxel is an extension of a pixel in the 2D plane to the 3D space. First, divide the 3D space into dense grids. Second, fill a grid with a voxel (like a cube) if the proportion occupied by objects in the scene exceeds a customized threshold; otherwise, the grid is void. Voxels can be recorded with a binary value ("1" for exist, "0" for not exist), sorted by the grid position. The color of each voxel

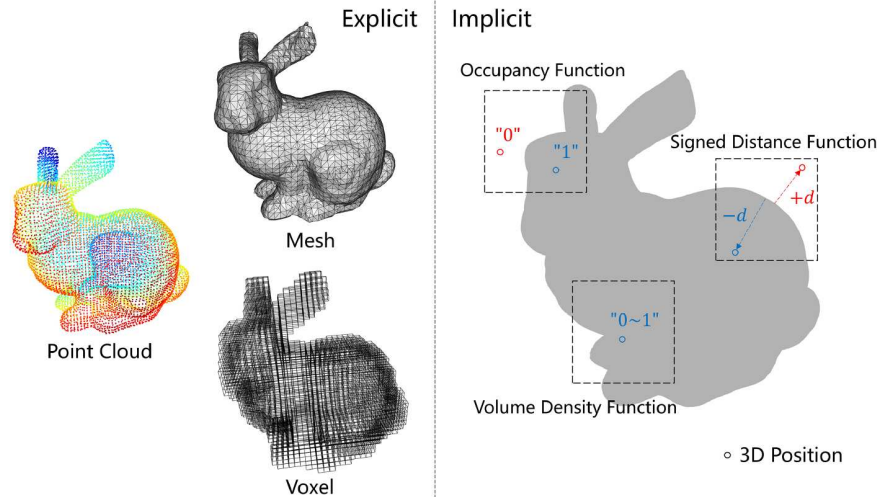


Figure 1 (Color online) 3D data representations for Stanford bunny (in implicit representation, color red stands for “outside”, color blue stands for “inside”, d is the minimum distance to the surface).

can be obtained from the color of the original object at that position. Voxel is convenient for constructing and editing, and can inherit methods in 2D image processing, such as convolution. However, there is a dilemma: the denser the grid, the more refined the representation, but the greater the storage and computation burden. Meshes fit surface in the scene with polygons (usually triangles or quadrilaterals) stitched together, which can be recorded by the vertices of each polygon in a row. The texture of the whole scene is mapped by an additional image file. Mesh is currently the most widely used form in 3D industrial and artistic design, with the advantages of intuitive, clear point-line-plane logic and sophisticated support for processing and rendering. Same as in voxel, the dilemma remains. Due to the limitation of resolution when displaying, a 3D point at any location can be interpolated through the closest several vertices around.

Implicit representations include occupancy function, signed distance function, and radiance fields. Occupancy function continuously encodes every position of the scene with a binary value. If this position is occupied by objects in the scene, “1” is recorded. Otherwise, “0” is recorded. Signed distance function (SDF) continuously encodes every position of the scene with a value that measures the minimum distance to any object’s surface. If the examined position is outside the object, the value is taken as positive, and if inside the object, it is taken as negative. Connecting all the points with a value of “0” gives the object’s surface, which is called a zero-level set. Considering the simplicity and practicability of operation, the SDF values of positions far from the surface are usually not required. The truncated SDF (TSDF), which only calculates the distance values within a certain range, is proposed. Radiance fields, or volume density function, continuously encode every position of the scene with a density value, assuming that the space is filled

with particles or “fields” (imagine as electromagnetic fields). From a statistical perspective, the greater the density value, the higher the likelihood that the position is occupied by an object. Fields are translucent, and what we see is the superposition of all fields along the ray of view. Observing from different directions results in different outcomes even towards the same position. This benefits the expression of optical reflection and refraction, leading to a realistic rendering output as shown in NeRF. The advantages of implicit representation are that it has infinite resolution, requires low storage space, and can be optimized through learning-based methods such as neural networks. The disadvantages include visual inaccessibility and the difficulty of integrating with common applications. 3D Gaussians aim to populate scenes with numerous ellipsoids of varying 3D positions, sizes, and orientations. The volume density at any point within an ellipsoid is computed by a 3D Gaussian function, while the density at any position in the overall space is determined by the combined contribution of all Gaussians that cover it. To be more exact, the 3D Gaussian approach is a hybrid explicit-implicit form of representation.

For practical needs, different representation formats can be converted to each other. Taking mesh as an example, the conversions are commonly achieved through methods including Poisson surface reconstruction [32], marching cubes algorithm [33,34], and density transformation [35,36], as shown in Figure 2.

3 Modern 3D reconstruction: NeRF and 3DGS

The groundbreaking work in modern 3D reconstruction was NeRF [30] in 2021, and another milestone was 3DGS [31] in 2023. Although NeRF and 3DGS differ in their specific

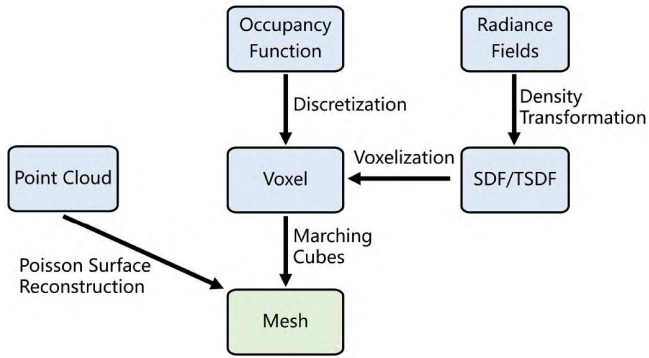


Figure 2 (Color online) Different 3D representations converted to mesh with methods commonly used.

representation methods, they both achieve visually smooth transitions by describing the scene through continuous occupancy modeling, as shown in Figure 3. The observed results are the integral of all radiance fields (the smallest representing units to construct the 3D scene) along the ray of view, thus offering greater completeness and flexibility in high-quality rendering. Owing to the photorealistic output, NeRF and 3DGS have attracted widespread attention in academia and industry, forming a relatively independent area. To facilitate a macroscopic understanding, we prefer to introduce these two methods using a comparison method distinguished from the existing literature, as shown in Table 1.

Origin: NeRF originates from the idea of plenoptic function [37], which describes the intensity of the lights around us. As a simpler version, NeRF uses a continuous 5D function that describes the radiance emitted in each direction at each position. 3DGS derives from the point sample rendering [38], which reconstructs the real scene with the basic unit of 3D points or point primitives like circulars and ellipsoids.

Representation and parameters: NeRF represents the scene with a continuous density field, which is an implicit form.

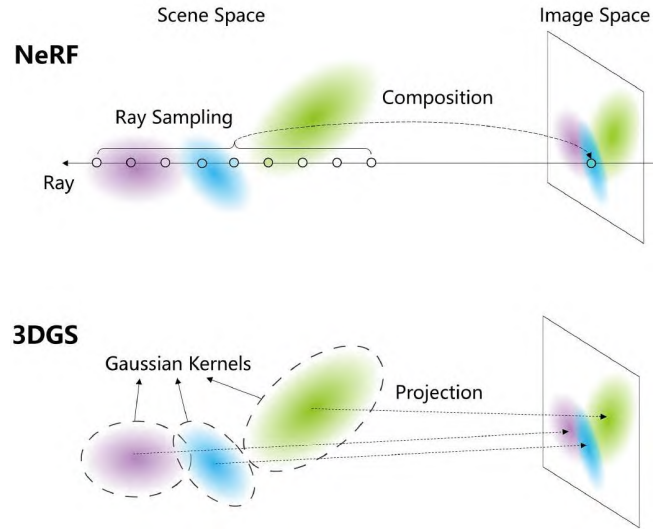


Figure 3 (Color online) NeRF and 3DGS.

Microscopically, the field is composed of particles with a certain density (or opacity, depending on their position) and color (depending on both position and direction). 3DGS represents the scene with 3D Gaussians, which is an explicit form. Each Gaussian kernel is an ellipsoid parameterized with position, orientation, scale, opacity, and color. The first three parameters are represented by a covariance matrix. The opacity value decays from the center of the ellipsoid to its edge based on a Gaussian function. The color value is encoded separately by spherical harmonics.

Input: NeRF and 3DGS both require images with camera parameters, which can be estimated via traditional SfM. A sparse point cloud is also needed in 3DGS for the initialization of Gaussian kernels.

Model to record 3D features: NeRF records spatial information in a multilayer perceptron (MLP). The optimization of network parameters is actually the optimization of the density distribution. 3DGS fits the density distribution by

Table 1 Comparison of NeRF and 3DGS

	NeRF	3DGS
Origins	Plenoptic function	Point sample rendering
Representation	Particles (color, opacity)	Gaussian kernels (position, orientation, scale, opacity, and color)
Input	Images with camera parameters	Images with camera parameters, sparse point cloud
Model	Multilayer perceptron	Density control
Loss	Pixel-wise loss	Photometric loss
Rendering	Volumetric rendering	Point-based rendering
Projection	NDC	EWA+NDC
Output	Rendered images	Rendered images
Performance	High render quality Slow training and rendering Lower storage requirements	High render quality, weaker in complex lighting Fast training and real-time rendering High storage requirements

cloning and splitting the Gaussian kernels, which can be adaptively controlled during optimization.

Loss function: Both NeRF and 3DGS calculate the loss of the model by comparing the input and output images with the same view perspectives. However, NeRF conducts the propagation according to the color of related pixels. 3DGS conducts the propagation based on the appearance and structure similarity of each image correspondence. Different definitions of loss functions are due to different rendering schemes.

Render scheme: NeRF renders every pixel in the image by compositing density along the ray of view, which is a volumetric method. 3DGS renders the entire image by directly splatting Gaussian kernels within the field of view into the image plane, which is a point-based method. They both employ α -blending for the intersection of density fields.

Projection model for image formation: NeRF and 3DGS both transform the rays from camera space to normalized device coordinate (NDC), which preserves the parallel lines. Then, a common rasterization pipeline in computer graphics can be applied for image formation. 3DGS first transformed the Gaussian kernels in camera space to ray space by elliptical weighted average (EWA) [39], which introduces a local affine approximation during projection.

Output: Both NeRF and 3DGS render images with the same view as the source image dataset during the training process, and synthesize novel view images in prediction.

Performance: In terms of reconstruction quality, both NeRF and 3DGS produce photorealistic novel views. 3DGS is less effective in modelling complex lighting and reflections. In terms of computational efficiency, the original NeRF often takes hours to days for training and tens of seconds to render a single frame; 3DGS can complete training within minutes and achieve a rendering speed of over 200 FPS. In terms of storage requirements, NeRF is compact in a few MBs for its implicit representation; 3DGS requires a few GBs to store millions of Gaussians.

Subsequent improvements for NeRF mainly focus on training and rendering efficiency, scalability, dynamics, and material-lighting decoupling. Mip-nerf [40] achieves an anti-aliasing rendering output by shifting the sampling process from rays to cones, and Mip-nerf 360 [41] uses a non-linear scene parameterization to “warp” infinite space into a finite bounding box, allowing the network to allocate more capacity to nearby content than to empty areas. Neural sparse voxel fields (NSVF) [42] is a hybrid explicit-implicit architecture that addresses the efficiency limitations of the original NeRF by combining voxel grid and neural features. Furthermore, Instant-neural graphics primitives (NGP) [43] dramatically reduces the training time to seconds or minutes through multiresolution hash encoding.

For large-scale scene reconstruction, Mega-NeRF [44] leverages geometric clustering to partition training images or

pixels into different NeRF submodules, which can be trained in parallel. Block-NeRF [45] divides large urban scenes into smaller overlapping blocks, each represented by an independently trained NeRF. These blocks are dynamically combined during rendering. BungeeNeRF [46] uses a progressive training strategy that incrementally adds details as the viewpoint zooms in, namely from satellite-level overviews to ground-level details. For temporal dynamic reconstruction, Dynamic NeRF [47] introduces a deformation field that maps points from any time step back to a canonical (static) space. Nerfies [48] focuses on modelling non-rigid deformations, particularly for scenes like human faces or bodies. A hypernetwork architecture is introduced to generate weights for a NeRF based on a latent code for deformation. For materials and lighting decoupling, NeRFactor [49] uses a novel visibility network to approximate how much light from any direction reaches a surface point. Neural reflectance decomposition (NeRD) [50] takes a more integrated approach by baking lighting into a neural representation during volume rendering. NeRF-Texture [51] disentangles a scene into a smooth underlying base shape and the detailed latent features on it. It allows for the texture’s rich, view-dependent appearance to be learned separately from its geometry. Subsequent improvements for 3DGS are also carried out mainly in the aforementioned aspects. Adaptive radius (AdR)-Gaussian [52] uses adaptive radius early culling and axis-aligned bounding box (AABB) culling to accelerate the rendering process. FlashGS [53] designs a compute unified device architecture (CUDA) library for 3DGS kernel optimizations at the low level. GSCore [54] directly builds a dedicated GPU hardware specifically designed for the 3DGS rendering pipeline. Instant-Splat [55] and SfM-free Gaussian splatting (SFGS) [56] both eliminate the need for traditional SfM, which are often slow and unreliable in sparse-view scenarios.

One approach to improve reconstruction quality and reduce storage cost is replacing the original Gaussian ellipsoids with other flexible kernel shapes [57–61]. Deformable radial kernel (DRK) splatting [58] replaces fixed Gaussian primitives with learnable, planar kernels. Each DRK primitive uses multiple radial bases with adjustable angles and scales, a hybrid L1/L2 distance metric, and an adaptive sharpening function to flexibly model diverse shapes, including sharp edges and curved surfaces. Exact volumetric ellipsoid rendering (EVER) [61] performs exact volume rendering with constant-density ellipsoids, eliminating popping artifacts and view-dependent density. Built on ray tracing, it supports effects like defocus blur and fisheye distortion. Another approach is enhancing 3DGS through the incorporation of texture fields or distillation from NeRF. Radsplat [62] uses a pre-trained radiance field as both an initialization prior and a clean supervision signal to guide the optimization of a point-based Gaussian representation,

leading to more stable training and higher-quality results. Viewing direction Gaussian splatting (VDGS) [63] uses 3D Gaussians to represent scene geometry and a NeRF-based small neural network to adjust opacity based on viewing direction. This enables better modeling of reflections, transparency, and shadows without sacrificing 3DGS's efficiency. Neural shell texture (NeSTS)plating [64] proposes a novel geometry-appearance disentangled representation with fewer 2D Gaussian primitives and a compact implicit appearance model inspired by Instant-NGP.

For large-scale scene reconstruction, Vast-Gaussian [65] partitions the large scene into multiple manageable, overlapping tiles. Each tile is reconstructed independently and then merged coherently. Hierarchical 3DGS [66] and LoD Gaussian splatting with efficient rendering (LODGE) [67] generate the scene in a coarse-to-fine manner based on a level-of-detail (LoD) strategy. Moreover, 4D Gaussian [68,69], dynamic 3D Gaussians [70], and LODGE add an explicit temporal dimension for Gaussian parameters to meet the requirements of dynamic expression. Deformable 3D Gaussian [71], GauFR [72], and sparse-controlled Gaussian splatting (SC-GS) [73] learn a set of Gaussians in a canonical space and a deformation field (often an MLP) that maps these Gaussians to their specific state at any given time. For materials and lighting decoupling, Relightable 3D Gaussian [74] associates each Gaussian with normal direction, BRDF parameters, and incident lighting. Spatially-varying Gaussian splatting for inverse rendering (SVG-IR) [75] extends original Gaussians to ones with spatially varying material attributes and normals, achieving an inverse rendering pipeline.

Diffusion Gaussian splatting combines diffusion models with 3DGS for high-fidelity 3D reconstruction and generation. In this context, diffusion models are used to generate and refine 3D Gaussian representations, enabling the efficient production of 3D assets even with a limited number of input images. Gaussian splatting diffusion (GSD) [76] is a single-view 3D reconstruction method focused on inference from 2D to 3D. It leverages view-guided sampling, using the differentiable splatting renderer to backpropagate image-space gradients and refine the 3D Gaussians during the denoising process, ensuring high-fidelity geometry and novel view synthesis without requiring model fine-tuning. An auxiliary 2D diffusion model further polishes rendered views to iteratively enhance the 3D reconstruction quality. DiffGS [77] is a general-purpose and functional 3D content generation model, which can be applied to various types of input in different generation tasks. It proposes a diffusion-based generative model for 3DGS that overcomes the challenge of its unstructured nature by representing Gaussians via three continuous, disentangled functions: one for geometry (probability), one for color, and one for transformations (scale, rotation, opacity). An octree-guided sampling and

optimization algorithm is introduced to extract high-quality Gaussians at arbitrary densities, enabling efficient unconditional and conditional 3D generation.

The review literature on NeRF includes refs. [78–80]. Nerfstudio [81] also provides a modular framework for the implementation of relevant methods. The review literature on 3DGS includes refs. [82–85]. Gaustudio [86] also provides a modular framework for the implementation of relevant methods. As a supplement to the existing summaries that primarily focus on basic principles and specific improvements, the latest works on geometry extraction based on radiance fields representation will be discussed in the next section.

4 Modern 3D reconstruction for geometry extraction

In current applications like industrial manufacturing, urban planning, film and television, games, the format of 3D data mostly required is meshes, which are sufficiently supported by 3D modelling and rendering engines. Therefore, geometry extraction remains a basic and critical issue in 3D scene reconstruction. However, both NeRF and 3DGS originally aim at novel view synthesis (NVS), in which the representation is optimized based on photometric supervision. To meet the demand for geometry, researchers have carried out a series of works, which will be discussed below. The timeline of related works is shown in Figure 4.

4.1 NeRF-based geometry reconstruction

NeRF uses an implicit 3D representation, which is not convenient for geometry extraction. A middleware is usually introduced for surface finding. As a pioneer, unifying neural implicit surfaces (UNISURF) [87] aims to render surfaces and volumes without a mask. A coarse-to-fine strategy is introduced. First, the surface is estimated via the occupancy field, which is calculated through a binary classifier. Then, an interval around the estimated position along the ray is defined to capture a more accurate geometry.

NeuS [35] and VolSDF [36] firstly map each 3D coordinate to an SDF output. In NeuS, the SDF values are then converted to volume densities via the logistic density distribution. Additionally, an unbiased weight function is derived to prioritize the first surface intersection during ray integration, avoiding occluded region artifacts. VolSDF calculates density values through Laplace's cumulative distribution function (CDF). Then, a rectangle ruler is introduced to bound the opacity approximation error. This bound can improve the efficiency of volume sampling and surface finding. As an improved version, NeuS2 [88] constructs a framework connecting an SDF network with an

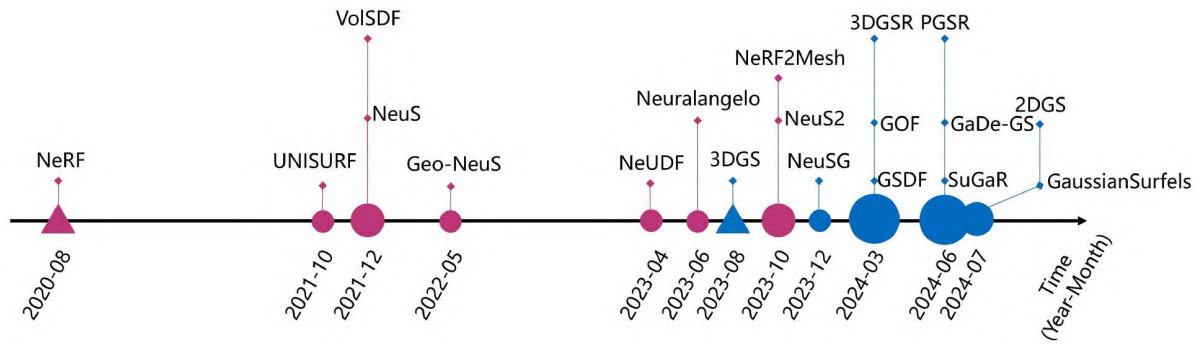


Figure 4 (Color online) Timeline of modern 3D geometry reconstruction (red: NeRF-based; blue: 3DGS-based).

RGB network. The volume feature is encoded by multi-resolution hash encodings. To avoid computational overhead, a second-order derivative of the ReLU-based MLP is applied to loss backpropagation during optimization. According to the fact that the sign of the SDF reverses once an intersection occurs, Geo-NeuS [89] discusses the sampling error when SDF is applied to surface estimation. Moreover, photometric consistency in MVS is introduced to supervise the extraction of the implicit surface. Inspired by Instant-NGP, Neuralangelo [90] incorporates the representation power of multi-resolution hash encoding into neural SDF representations. Higher-order derivatives are calculated to handle the locality problem during gradient propagation. The coarse-to-fine optimization strategy is applied to achieve a progressive reconstruction. Adopting the coarse-to-fine strategy, NeRF2Mesh [91] performs a joint optimization of geometry and appearance. In the first stage, a coarse mesh is estimated by applying the Marching Cubes algorithm to the density field. In the second stage, a fine mesh is calculated through adaptive adjustment of the face density inspired by the iteratively reweighted least squares algorithm [92]. NeUDF [93] firstly leverages the unsigned distance function (UDF) to represent complex geometries with arbitrary topologies, including complex shapes with open boundaries. A new weighting function tailored for UDF is introduced to avoid biased rendering and artifacts. A dedicated importance sampling strategy is proposed to balance gradient regularization across the surface.

4.2 3DGS-based geometry reconstruction

3DGS uses an explicit 3D representation, namely, Gaussian kernels, which directly fill the scene. The key problem lies with the surface estimation of the anisotropic superposition of the Gaussians within a specific field of view.

Combining implicit surfaces and 3DGS, NeuSG [94] builds a unified framework and performs a joint optimization. Gaussian kernels are flattened according to the minimal scaling factor while aligned with surface normals predicted by the aforementioned NeuS. Meanwhile, NeuS refines im-

plicit surfaces using 3DGS-generated point clouds. More directly, Gaussian splatting meets SDF (GSDF) [95] introduces a dual-branch architecture that combines 3DGS with SDF. Derived from Scaffold-GS [96], the GS-branch is used to guide the ray sampling process in the SDF-branch. In turn, the SDF-branch is used to guide the density control of the Gaussians. Moreover, a mutual geometry consistency between them is considered. SuGaR [97] introduces a regularization term to estimate the difference between the ideal SDF and the one produced from current Gaussians. Another regularization term is introduced to estimate the normal difference. As a result, the Gaussian kernels are encouraged to align more closely with the surfaces in the scene.

To improve the surface alignment, GaussianSurfels [98] flattens the original Gaussians into Gaussian surfels by setting the z -factor of the scale matrix to zero. The depth and normal of each pixel in the images are calculated through the splatting and α -blending process. Depth, normal, and their consistency are considered in the optimization. More directly, 2DGS [99] replaces 3D Gaussians with 2D Gaussians to represent the scene geometry, which brings the benefit of multi-view consistency. A ray-splat intersection method is introduced to conduct the projection of the elliptical slice to the image space. Similarly, planar-based Gaussian splatting for surface reconstruction (PGSR) [100] flattens each 3D Gaussian into a disk by minimizing the minimum factor in the scale matrix. Then the normal of each Gaussian is estimated naturally. As a result, depth can be determined through the projection relationship. Moreover, multi-view photometric and geometric regularization are introduced to maintain consistency.

Derived from ray tracing-based volume rendering, GOF [101] presents a Gaussian opacity fields that estimate the position where each Gaussian kernel reaches its maximum opacity along the ray. With GOF, surfaces are directly extracted by identifying their level sets, like in UNISURF. 3DGSR [102] estimates the opacity of each Gaussian according to the distance to the implicit surface calculated by SDF. In turn, the geometry attributes of Gaussians regularize the SDF. Depth and normal are also accumulated along the

ray through α -blending. Rasterizing depth (RaDe)-GS [103] computes the depth map and normal map for Gaussian splats using a rasterization method. The intersection points of the rays and Gaussians in the field of view are transformed from the camera space to the ray space. Therefore, the final depth is accumulated considering the opacity. Meanwhile, similar operations are performed on the normals.

4.3 Methodology summary

To establish a standardized research framework and inspire subsequent work, we propose the following fundamental strategies for geometry extraction, derived from a systematic analysis of the existing literature.

Threshold strategy: Based on the hypothesis that the density value only depends on position and is independent of the direction of view, a threshold can be applied to the density of the field generated by NeRF or 3DGS. A position is occupied if its density exceeds the threshold. However, the output is highly likely to be filled with noise, since the original modern 3D reconstruction is photometrically supervised.

GOF strategy: Estimate the intersection or occlusion with the geometry surface along the camera ray according to the density distribution, as shown in Figure 5. The density distribution is estimated by the α -blending of each unit with its maximum density along the ray [101].

Distance function (DF) strategy: Transform the density distribution into a signed or unsigned distance function, referred to as SDF and UDF, respectively. Then the surfaces in the scene can be extracted through the zero-level set. For example, the SDF strategy is shown in Figure 6. Finally, the DF is optimized through the backpropagation of the density loss. The key problem is to find an unbiased transformation [35]. Further optimization involves the incorporation of the eikonal term, curvature term, depth term, and normal term in the following regularization strategy.

Flattening strategy: Represent the scene directly with a 2D unit that is intuitive for surface extraction, as shown in Figure 7. For example, the 2D version Gaussian is like an elliptical slice or a surfel [99]. The 2D unit is initialized

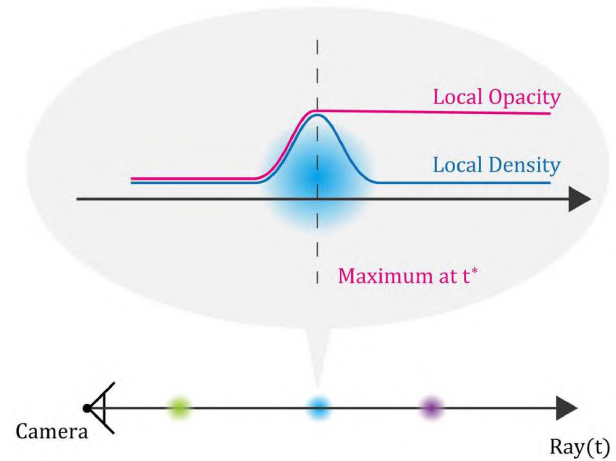


Figure 5 (Color online) GOF Strategy (blue line: density for a Gaussian in local space; pink line: opacity or density accumulated along the ray) [101].

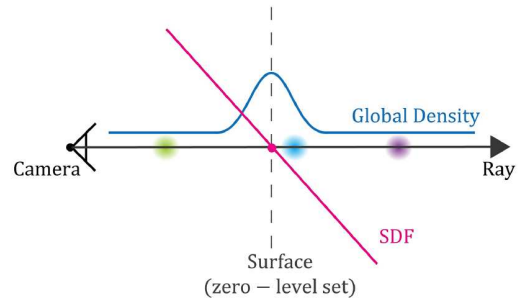


Figure 6 (Color online) SDF Strategy (blue line: density distribution in global space; pink line: signed distance function used for surface finding) [35].

either according to the point cloud from SfM or based on a 3D unit with one-dimensional attenuation. The corresponding regularization is the flattening term as defined in eq. (11).

Regularization strategy: Design appropriate regularization terms for the loss function. Regularization encourages the extraction of surface or volume while keeping the photometric correction. It is back-propagated to the model for optimization.

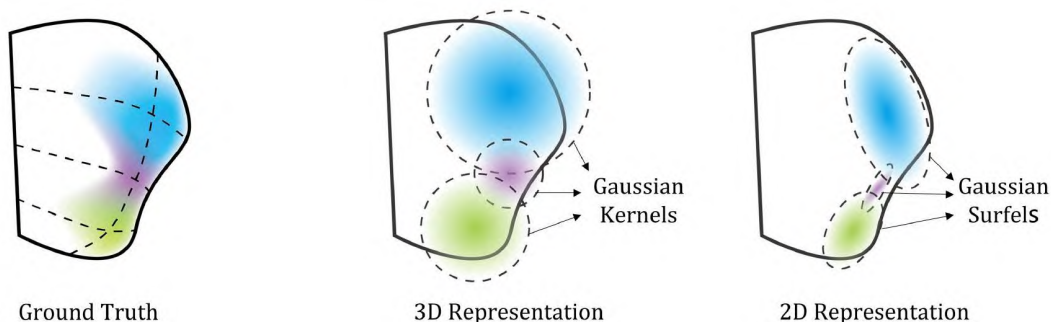


Figure 7 (Color online) Flattening strategy.

Common regularization terms are listed below.

Color term:

$$\mathcal{L}_{\text{color}} = \frac{1}{n} \sum_k \|(\hat{C}_k - C_k)\|, \quad (1)$$

which computes the RGB value difference between the rendered $\hat{C}_k$ and the ground truth C_k . k is the index of sampling rays in NeRF or index of images in 3DGS. n is a normalization term. $\|\cdot\|$ denotes the operation to obtain the normal form.

D-SSIM term [104]:

$$\mathcal{L}_{\text{SSIM}} = \frac{1 - \text{SSIM}(x, y)}{2}, \quad (2)$$

where $\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$, in which μ_x

and μ_y , σ_x and σ_y , and σ_{xy} are mean, standard deviation, and covariance computed within a local square window, respectively; C_1 and C_2 are constants introduced to prevent unstable output.

Eikonal term [105]:

$$\mathcal{L}_{\text{eik}} = \frac{1}{n} \sum_i (\|\nabla f(\mathbf{x}_i)\|_2 - 1)^2, \quad (3)$$

where $f(\mathbf{x}_i)$ is the SDF of the 3D position $\mathbf{x}_i$. The eikonal term encourages the gradient of SDF in the direction the distance field changes most steeply.

Mask term:

$$\mathcal{L}_{\text{mask}} = \text{BCE}(M_k, \hat{O}_k), \quad (4)$$

where $\text{BCE}(\cdot)$ is the binary cross-entropy loss. $\hat{O}_k$ is the density accumulated. M_k is an optional step prior to separating the foreground from the background.

Curvature term:

$$\mathcal{L}_{\text{curv}} = \frac{1}{N} \sum_i |\nabla^2 f(\mathbf{x}_i)|, \quad (5)$$

which computes the mean curvature (second-order gradients) of SDF for N sampled points. The curvature term encourages the smoothness of the reconstructed surface.

Depth term:

$$\mathcal{L}_{\text{depth}} = \frac{1}{N} \sum_i |\hat{f}(\mathbf{x}_i) - f(\mathbf{x}_i)|, \quad (6)$$

where $\hat{f}(\mathbf{x}_i)$ is an estimated SDF/depth, $f(\mathbf{x}_i)$ is the ideal SDF/depth. Depth term encourages the reconstructed geometry to concentrate on the a priori surface.

Normal term:

$$\mathcal{L}_{\text{norm}} = \frac{1}{N} \sum_i \left\| \frac{\nabla f(\mathbf{x}_i)}{\|\nabla f(\mathbf{x}_i)\|_2} - n_g^* \right\|^2, \quad (7)$$

where n_g^* is the normal of the Gaussian with the largest

contribution at position $\mathbf{x}_i$.

Depth-normal consistency term:

$$\mathcal{L}_{\text{con}} = 1 - \tilde{\mathbf{N}} \cdot N(V(\tilde{\mathbf{D}})), \quad (8)$$

where $\tilde{\mathbf{N}}$ and $\tilde{\mathbf{D}}$ are normal and depth of each pixel calculated through α -blending. $V(\cdot)$ transforms each pixel with its depth to a 3D point, and $N(\cdot)$ calculates the normal from neighboring points using the cross product.

Volume term [106]:

$$\mathcal{L}_{\text{vol}} = \sum_{i=1}^{N_g} \text{Prod}(s_i), \quad (9)$$

where N_g denotes the number of Gaussians in the scene, $\text{Prod}(\cdot)$ is the product of a vector, s_i is the scale vector of each Gaussian. Volume term encourages the Gaussians to be small with minimal overlapping.

Densification term:

$$\mathcal{L}_{\text{den}} = e^{-(o_i - 0.5)^2 / 0.05}, \quad (10)$$

which pushes each Gaussian's opacity o_i either close to 1 or close to 0 with a sigmoid function, encouraging the condensation of surfaces. This can also be achieved through an entropy regularization.

Flatten term:

$$\mathcal{L}_{\text{flat}} = \|\min(s_1, s_2, s_3)\|_1, \quad (11)$$

which minimizes the minimum factor of the scale matrix $S = \text{diag}(s_1, s_2, s_3)$ for a Gaussian. $\text{diag}(\cdot)$ denotes a diagonal matrix. Compared with the aforementioned flattening strategy, which directly introduces the 2D hypothesis, the flattening term is a softer method.

Multi-view consistency term (including geometric and photometric):

$$\mathcal{L}_{\text{mvgeom}} = \frac{1}{N} \sum_{\mathbf{p} \in V} \|\mathbf{p} - \mathbf{H}_{nr} \mathbf{H}_m \mathbf{p}\|, \quad (12)$$

$$\mathcal{L}_{\text{mvrgb}} = \frac{1}{N} \sum_{\mathbf{p} \in V} (1 - \text{NCC}(\mathbf{I}_r(\mathbf{p}), \mathbf{I}_n(\mathbf{H}_m \mathbf{p}))), \quad (13)$$

where $\mathcal{L}_{\text{mvgeom}}$ describes the forward and backward projection error of each pixel $\mathbf{p}$ in the reference image set V . N is the number of pixels in the reference image. $\mathbf{H}_{nr}$ and $\mathbf{H}_m$ are homography matrices between the reference image and a neighboring frame. $\mathcal{L}_{\text{mvrgb}}$ describes the normalized cross-correlation ($\text{NCC}(\cdot)$) of pixel RGB value $\mathbf{I}_r(\mathbf{p})$ and $\mathbf{I}_n(\mathbf{H}_m \mathbf{p})$ in the reference and neighboring frame, respectively. The multi-view consistency term continues the reprojection idea in the MVS, which ensures the consistency of reconstructions from different view perspectives.

In summary, the strategies used in each method for geometry extraction are shown in Table 2. What should be noted is that the regularization terms above are basic examples.

Table 2 Common strategies used for geometry reconstruction

Methods		Strategies															
		Threshold	GOF	DF	Flattening	Regularization											
						$\mathcal{L}_{\text{color}}$	$\mathcal{L}_{\text{SSIM}}$	$\mathcal{L}_{\text{cik}}$	$\mathcal{L}_{\text{mask}}$	$\mathcal{L}_{\text{curv}}$	$\mathcal{L}_{\text{depth}}$	$\mathcal{L}_{\text{norm}}$	$\mathcal{L}_{\text{con}}$	$\mathcal{L}_{\text{vol}}$	$\mathcal{L}_{\text{den}}$	$\mathcal{L}_{\text{flat}}$	$\mathcal{L}_{\text{mvgeom}}$
NeRF-based	UNISURF [87]	✓				✓							✓				
	NeuS [35]			✓		✓		✓	✓								
	VolSDF [36]			✓		✓		✓									
	Geo-NeuS [89]			✓		✓		✓			✓						✓
	NeRF2Mesh [91]	✓				✓				✓					✓		
	NeuS2 [88]			✓		✓		✓									
	Neuralangelo [90]			✓		✓		✓		✓							
	NeUDF [93]			✓		✓		✓	✓								
3DGS-based	SuGaR [97]			✓		✓	✓				✓	✓				✓	
	NeuSG [94]			✓		✓	✓	✓					✓			✓	
	GSDF [95]			✓		✓	✓	✓		✓	✓	✓		✓			
	2DGS [99]				✓	✓	✓				✓	✓					
	GOF [101]		✓			✓	✓				✓	✓					
	GaussianSurfels [98]				✓	✓	✓		✓			✓	✓		✓		
	3DGSR [81]			✓		✓	✓	✓			✓	✓			✓		
	RaDe-GS [82]		✓			✓	✓				✓	✓					
	PGSR [79]					✓	✓					✓				✓	✓

The specific form of the regularization may vary in different methods. The NeRF-based geometric reconstructions mostly use DF as a bridge to concretize the vague radiance fields, thereby achieving surface finding. Among them, the best experimental results on the DTU dataset (e.g., Neuralangelo) achieve a chamfer distance (mm) of above 0.60, but require more than 12 h of computation time. In contrast, 3DGS adopts an explicit scene representation, which is more efficient in geometry extraction. For example, PGSR achieves a chamfer distance (mm) on the DTU dataset under 0.50 in 1 h; 2DGS can achieve a result of 0.8 in just half an hour. Therefore, the latest radiance field-based geometry reconstruction work mainly takes the 3DGS-based approach. Improving reconstruction accuracy and speed remains an ongoing pursuit in the field.

Another important issue is that, to achieve smoother and less noisy scene geometry, relevant geometric constraints or regularization terms are usually introduced, which more or less interfere with the existing photometric loss [96,99]. As a result, in most geometry reconstruction works, the accuracy of rendering results decreases. Logically, for one scene, the expression should be consistent in either graphic views or geometric views, and there should be an enhancing effect between the two. Building a reconstruction framework with a dual branch for both appearance and geometry has become a solution to enhance overall reconstruction quality. For instance, in works such as 3DGSR and GSDF, mutually optimizing both existing 3DGS and its corresponding SDF is

employed, achieving a certain balance between rendering quality and geometric accuracy. Meanwhile, geometry extraction can also be compensated for by NeRF-based appearance representations. In NeSTSplatting, Geometry is represented explicitly by 2D planar Gaussian surfels (as in 2DGS), which act as efficient samplers of surface structure. Appearance or texture is modeled globally using a continuous neural shell texture, implemented via a multi-resolution hash-encoded feature field inspired by Instant-NGP. This combination achieves high effectiveness and efficiency in both NVS and geometry reconstruction.

For the two basic tasks of 3D reconstruction, modern geometry extraction can integrate directly with existing technologies to support geometry creation and edition, especially for applications that require surfaces or models; modern novel view synthesis can efficiently render images from specific locations and directions according to the 3D representation model, especially for applications requiring visual output. Therefore, modern 3D reconstruction can feed traditional applications like industrial design and artistic creation with the convenience of object generation.

5 Future topics of modern 3D reconstruction

Modern 3D reconstruction adopts an architecture with low complexity and minimal requirements of source data. With the support of parallel computing powered by GPU, the ef-

iciency can be ensured, even reaching a real-time level. By detaching from the feature-based projection and optimization in traditional MVS, the versatile radiance field-based approach has pushed the boundary of 3D reconstruction. Currently, preliminary explorations are being carried out in potential directions that open up new avenues for solving bottleneck problems in areas like visual positioning and navigation, virtual reality, and light field display.

5.1 Modern 3D reconstruction in SLAM

Simultaneous localization and mapping (SLAM) aims to enable a mobile robot or an unmanned vehicle placed in an unknown environment to incrementally build a consistent map of this environment while simultaneously determining its location within this map [107]. Modern 3D reconstruction provides a new scene representation with the advantages of flexibility, completeness, and denseness. The camera pose and density map are jointly optimized with frames of sensor data incoming.

iMap [108], as the first NeRF-based SLAM, consists of two concurrent processes: tracking and mapping. In the tracking part, the camera pose is estimated based on the current RGBD frame and the implicit network. In the mapping part, the implicit network is optimized via differential rendering according to the RGB-D images of all key frames with camera parameters. Key frame is chosen if a frame has a low depth similarity to the ones that have been registered. Splat track and map (Spla-TAM) [109] first represents the map with explicit 3D Gaussians to increase computational efficiency via fast rasterization. This Gaussian-based SLAM system performs three main steps. Camera tracking: Camera pose of an incoming RGB-D frame is updated by minimizing the depth and color loss to the current map; Gaussian densification: New Gaussians are generated when the estimated depth of a position is much less than the ground truth; Map updating: The Gaussian map is adjusted through gradient-based optimization on depth and color within a limited number of key frames. Aimed at the basic but the hardest monocular SLAM problem, MonoGS [110] formulates camera tracking for the optimization of Gaussians. Both geometric and photographic regularization are introduced for an incremental 3D map reconstruction. This work achieves a 3 FPS live SLAM and can be seamlessly extended to an RGB-D setup. Another similar work is GS-SLAM [111], which employs the strategy of adaptive 3D Gaussian expanding mapping. Coarse-to-fine camera tracking and bundle adjustment are introduced in the optimization process. Structure-enhanced Gaussian splatting (SEGS)-SLAM [112] leverages a prior highly structured point cloud to initialize anchored 3D Gaussians to preserve geometric consistency and improve rendering fidelity. Camera pose is encoded into a latent space to model appearance variations based on ap-

pearance-from-motion embedding. OpenGS-SLAM [113] generates consistent pointmaps across frames, which store spatial relationships and scene geometry across multiple views. Adaptive scale mapper and dynamic learning rate adjustment are introduced for performance improvement. Voxel-based progressive Gaussian (VPGS)-SLAM [114] proposes a sequentialized voxel-based progressive 3D Gaussian representation for large-scale scenes. A 2D-3D fusion tracking method is introduced to combine the 3D Gaussian geometric information with 2D photometric information.

A general pipeline of radiance field-based SLAM is given in Figure 8. In the tracking process, the camera pose of a new incoming frame is estimated based on the current 3D Gaussian map according to photometric consistency. In the mapping process, the input frames are used to optimize the existing Gaussian map, and when a current frame is identified as a keyframe, newly generated Gaussians are incorporated to extend the Gaussian map. The performance of the related methods on the dataset Replica [115] and TUM RGB-D [116] is summarized in Table 3.

The survey [117] summarizes the evolution of SLAM, ranging from a hand-crafted way, through a deep learning-based approach, especially to radiance field-based methods. This comprehensive survey classified the SLAM based on NeRF or 3DGS into different categories according to the type of sensor used for data collection. Each category is further divided according to the way of geometric representation. The challenges and potential avenues are also discussed.

Modern 3D reconstruction provides dense localization and mapping output in both the constructing and optimizing steps of SLAM. However, compared with the traditional feature-based approach, a more precise and rich result calls for larger computational consumption, which is a challenge to the critical requirement of simultaneity in SLAM.

Relevant application domains based on SLAM are robotics and autonomous systems. During the process of sensing, denser and richer 3D data helps the understanding of the scene, such as in the tasks of recognition, localization, and segmentation [118]. Multi-sensor fusion is an effective approach currently, but efficient information extraction is hard from heterogeneous sensor data. One important reason is the lack of a unified and accurate 3D scene representation. Modern 3D construction is independent of the equipment used for data acquisition. The radiance fields can act as a flexible data fusor with less information loss.

Future work can be conducted in the following aspects: (1) Sparse and adaptive Gaussian management. Use view frustum culling, LOD, or occlusion culling to render only visible/relevant Gaussians per frame. (2) Hardware-aware parallel optimization. Leverage CUDA or other GPU-accelerated frameworks for parallel Gaussian rendering and

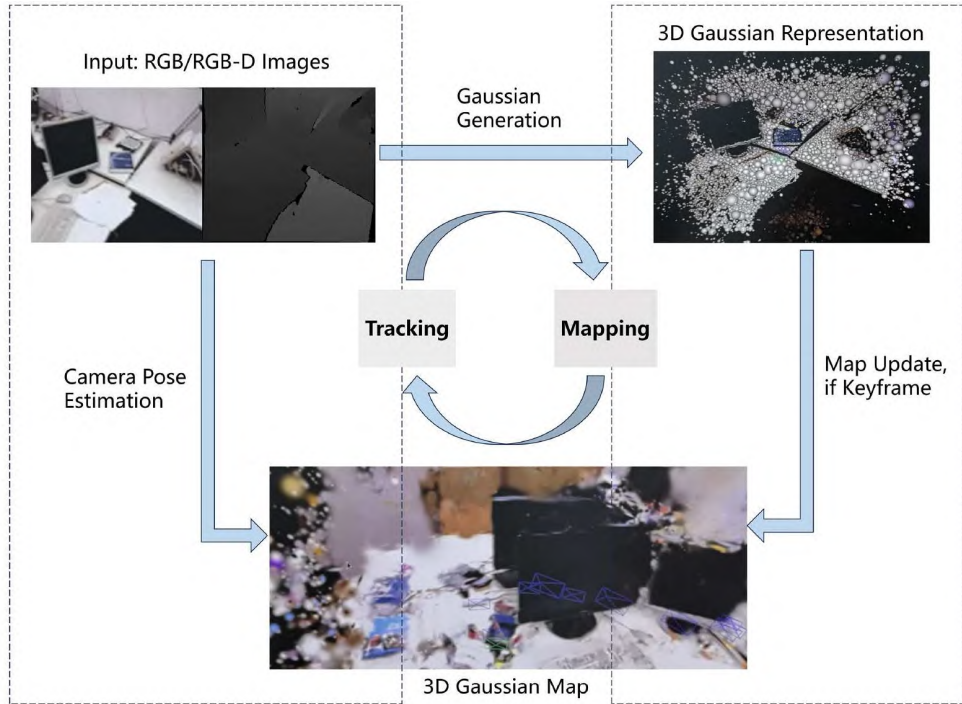


Figure 8 (Color online) The pipeline of radiance field-based SLAM.

Table 3 Performance of the GS-based SLAM methods on the dataset Replica and TUM RGB-D

Method	Replica (RGB-D)				TUM RGB-D (RGB-D)			
	Rendering			Camera tracking	Rendering			Camera tracking
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	RMSE $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	RMSE $\downarrow$
MonoGS [110]	38.94	0.97	0.07	0.58	24.37	0.80	0.23	1.47
GS-SLAM [111]	34.27	0.98	0.08	0.50	—	—	—	3.70
SplaTAM [109,112]	34.11	0.97	0.10	0.36	21.41	0.76	0.27	5.48
SEGS-SLAM [112]	39.42	0.98	0.02	0.43	26.03	0.84	0.11	1.53

gradient computation. Optimize memory access patterns and kernel launches. Decouple tracking, mapping, and rendering into separate threads or GPU streams for asynchronous computation. (3) Hybrid representations. Use a lightweight geometric representation for camera tracking and pose estimation, while reserving 3DGS for high-quality rendering and mapping. Maintain multiple levels of Gaussian density: sparse for tracking, dense for visualization, and switch dynamically.

5.2 Modern 3D reconstruction in virtual reality

Virtual reality (VR) is an immersive display and interactive technology that transmits computer-generated binocular images or videos to head-mounted display devices, achieving stereoscopic vision and providing interaction interfaces. VR has been pursuing visual realism and natural interactivity since its invention. Existing 3D content generation mostly

relies on the modelling-based solution, which is time-consuming and requires experience. Moreover, the 3D assets suffer from limited quality in lighting and texture rendering. In contrast, modern 3D reconstruction can inherently output photorealistic images for VR without redundant geometry extraction. According to the latest outcome in NeRF and 3DGS, the demands of high frame rate and high resolution in VR can be satisfied.

Foveated NeRF [119] is based on the egocentric characteristics of VR scenes. 3D radiance fields in NeRF are transformed to a concentric spherical coordinate space, which is discretized by different radii. Gaze-contingent synthesis divided the FOV into fovea-, mid-, and entire levels according to eccentricity ranges, considering adaptive monoscopic and stereoscopic acuity. Latency-quality joint optimization is conducted in the spatial-temporal domain. VR-NeRF [120] aims at the capture process of NeRF-based content acquisi-

tion. A customized capture rig mounted with cameras facing in different directions is designed for plenoptic observation. The rig also provides six degrees of freedom, which is critical for the scene scale rendering in VR. VR-GS [121] performs preprocessing on the input images through a 2D segmentation model to get separate objects represented by Gaussians. Bounding mesh reconstruction and tetrahedralization are utilized to achieve a two-level embedding for deformation. It enables users to intuitively deform and interact with virtual objects in real-time up to 25FPS. 3DGS-Avatar [122] presents a method to create animatable human avatars from a single monocular video using deformable 3DGS. It achieves real-time rendering (over 50 FPS) and fast training (around 30 min) by combining linear blend skinning for body articulation with a lightweight MLP for non-rigid cloth deformation and view-dependent color, along with geometric regularization to handle unseen poses.

Modern 3D reconstruction provides a shortcut from scene capturing to view rendering. However, an additional process for edge awareness is also needed in order to provide interaction with different objects in the scene. In practice, a customized visual hypothesis or supplement is introduced for the system design. To adapt to general scenarios, a genetic framework is imperative for a wider application.

5.3 Modern 3D reconstruction in physical simulation

Physical simulation is a technique that uses computers to simulate physical phenomena. It involves creating mathematical models to approximate the behavior and processes of real-world physics. The physical entities in physical simulation are mostly constructed in traditional ways, usually represented in the form of a geometric mesh. As a result, the resolution of the object is often limited, not enough for simulation calculations. There are two solutions: one is to simplify the experimental subject or mathematical model within an acceptable level; the other is to conduct object construction and physical simulation separately, that is, the simulation calculation is carried out on an ideal object, and then the results are fed back to the virtual object [123–126].

Modern 3D reconstruction brings a continuous scene representation without resolution limitation. If simulation calculations can be conducted on objects parameterized as neural networks or Gaussian kernels, a more consistent, rapid, and precise simulation can be achieved. Therefore, the key issue lies in the support of physical operation on implicit objects. VR-GS applies position-based simulation [127] to Gaussian kinematics, achieving real-time interaction. Phys-Gaussian [128] achieves physics simulation and visual rendering via 3DGS without any geometry embeddings usually used in traditional ways. Gaussian kernels evolve to a time-dependent version based on mass and momentum conservation models in continuum mechanics. The position and

deformation parameters are encoded and calculated in the form of an Eulerian grid and Lagrangian particles. The integration of physics into 3DGS is seamless, which is critical for the principle of “What you See is What you Simulate (WS2)” [129].

To address the current simulation’s need for explicit objects, real-time deformation on mesh or mesh-based radiance fields is proposed. NeRF-Editing [130] performs user-controlled 3D shape deformation on static NeRF without re-training. It achieves this by establishing a correspondence between an extracted explicit mesh (which the user deforms) and the implicit NeRF representation, using a tetrahedral proxy mesh to propagate surface edits into a volumetric deformation field that bends rays during rendering. NeuMesh [131] represents a 3D scene using a mesh scaffold, where disentangled geometry and texture features are stored directly on the mesh vertices. This explicit mesh-based structure enables intuitive and precise editing operations. Mesh-based GS [132] integrates 3D Gaussians with an explicit mesh. The mesh guides Gaussian initialization, splitting, and deformation, enabling better geometric fidelity and visual quality. Gaussians are bound to mesh faces and updated according to mesh deformation gradients. An interactive tool allows users to manipulate the mesh directly, achieving real-time deformation at 65 FPS and high-fidelity renderings (PSNR: 33.43, SSIM: 0.968, LPIPS: 0.034) on NeRF-Synthetic Datasets. Dynamic Gaussian mesh [133] builds on 3DGS to model dynamic geometry and appearance, and introduces Gaussian-mesh anchoring to ensure uniform distribution of Gaussians, supporting topology changes, vertex tracking, and cross-frame correspondence. It maintains 10 FPS high mesh querying and comparable rendering quality (PSNR: 27.24) on a real scene dataset.

A general pipeline of radiance field-based physical simulation and virtual reality is given in Figure 9. Generally, there are two main approaches for implementation. One is a mesh-based approach. First, the target object is segmented from the Gaussian scene; then, a mesh is embedded to it; finally, deformation operations are performed on the mesh based on simulation or interaction results. This approach is more efficient for its local mesh-level operation. The other is a Gaussian-based approach, which directly deforms the Gaussian scene based on simulation or interaction result, followed by segmentation, geometry extraction, and rendering of the relevant target objects as needed. This approach offers greater completeness and accuracy for its global Gaussian-level operation.

Relevant application domains based on virtual reality and physics simulation include digital earth, digital twin [134,135], and the metaverse [136,137]. During the process of virtualization and digitization, more complete and precise 3D data helps enhance realism and immersion. Traditional scene representation, which is mostly geometry-based, has

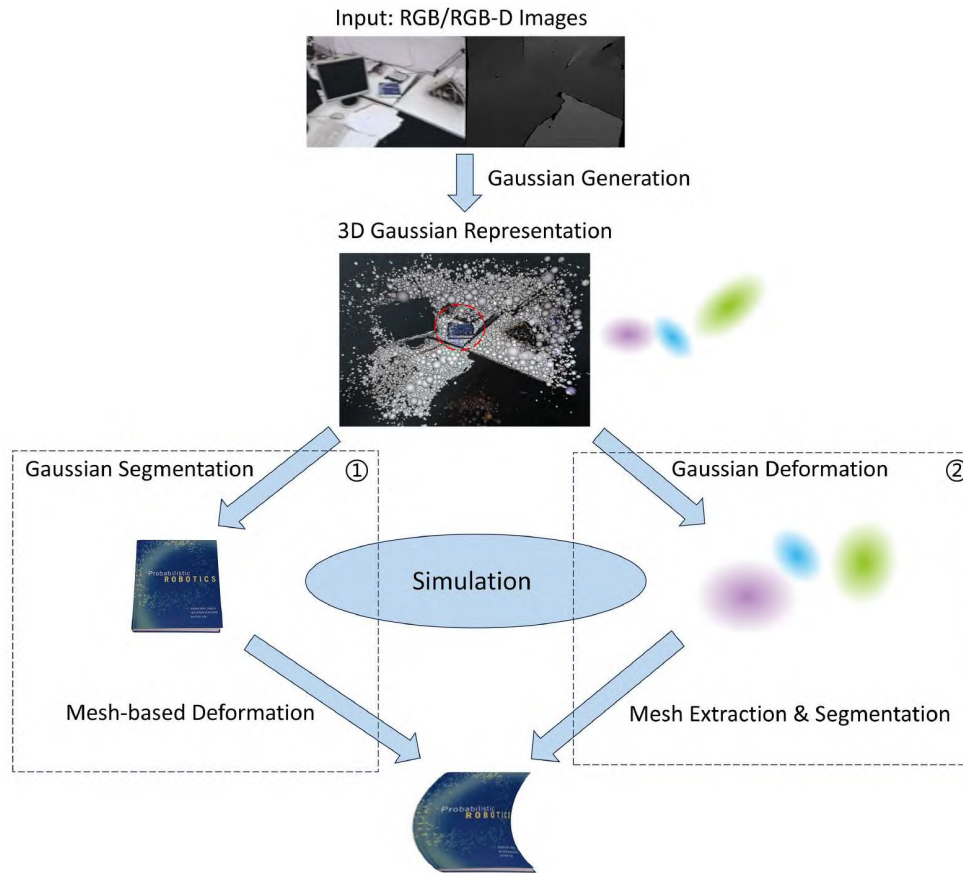


Figure 9 (Color online) The pipeline of radiance field-based physical simulation and virtual reality (① mesh-based approach; ② Gaussian-based approach).

an inevitable approximation and can hardly restore highly realistic textures, lighting, and shadow effects. Modern 3D reconstruction introduces the continuous representation, which makes precise simulation and timely feedback possible [138]. The photorealistic rendering and deformable modelling are important means to achieve effective interaction with the virtual scene.

Future work can be conducted in the following aspects: (1) Core representation and rendering pipeline. Implement highly optimized, GPU-accelerated splatting kernels that can handle the high resolution and dual-eye rendering required for VR headsets. (2) Physics-aware interactivity and deformation. Use a hybrid simulation-embedding architecture, like segment and cage, tetrahedral mesh embedding in VR-GS. (3) System integration and user experience. Combine all components (reconstruction, segmentation, inpainting, simulation, rendering, and generative AI) into a single, streamlined pipeline that can process new scenes in real-time. Design natural interactions (grabbing, scaling, throwing) that leverage the physics simulation.

5.4 Modern 3D reconstruction in light field display

Light field display (LFD) is committed to reconstructing the

light-ray distribution of the real 3D scene, including not only the intensity and color of the light, but also the direction of the light-ray [139,140]. LFD provides the depth cues in human vision without the need for special glasses or headsets, thereby avoiding discomfort, including the vergence-accommodation conflict [141] in conventional stereo displays.

Chen et al. [142] cut off the original NeRF for spatial sampling and encoding against low rendering speed. Furthermore, a coarse-to-fine 3D voxel rendering method is also introduced for acceleration. As a result, fast and precise view synthesis promotes the LFD process. Ma et al. [143] take advantage of the rendering efficiency of Instant-NGP. The light ray is encoded and input into the network, and the output serves as a projector array image in a multi-projector-type light-field display system. DirectL [144] believes that the multi-view rendering part is the major time-consuming. A common lenticular lens-based autostereoscopic display [145] consists of three steps: multi-view rendering, interleaving, and optical display. As a result, the multi-view images are encoded into an LCD panel. Then the light emitted from the pixels in the LCD panel is refracted by the lens to different spatial zones for perception. The ray-order rendering algorithm [146] is introduced for subpixel re-purposing. Since it inherently originates from the description

of light fields, using NeRF to acquire the resources required for LFD becomes a very natural approach. For 3DGS, an intermediate operation of ray-tracing is needed. This work achieves a 25 FPS rendering on a 48-viewpoint 2K resolution output. Shi et al. [147] integrate light field encoding algorithms with an optimized 3DGS rendering pipeline, enabling the direct generation of 8K-resolution light field images with 96 viewpoints in real time, approximately $30\times$ faster than traditional multi-view rendering methods, while maintaining high visual fidelity. Currently, the integration of radiance-based 3D reconstruction with LFD is still in its early stages, primarily limited to static light fields. Significant research opportunities remain in emerging areas such as dynamic or fluid light fields and interactive light fields. A general pipeline of radiance field-based light field display is given in Figure 10. After the Gaussian scene construction, novel view images are rendered according to the pixel encoding algorithm. The encoding pattern is calculated based on the optical parameters of the specific autostereoscopic display device. Finally, the light field images are composed of sub-pixels in

encoded multi-view images.

5.5 Modern 3D reconstruction in holography

Holography aims at the wavefront reconstruction of a real 3D scene based on wave optics, which is a potential solution for 3D display in the near future. Compared with light ray reconstruction (such as LFD), they share an equivalent concept but perform different procedures [140]. In recent computer-generated holography (CGH), a computer calculates the holographic fringe pattern and sets a spatial light modulator (SLM). Then the SLM diffracts the read-out light wave to yield the desired optical wavefront [148]. Because the wavefront propagation starts from the surface of objects extracted by multi-view images, one of the bottleneck issues lies in the generation of realistic images, which is the area modern 3D reconstruction just excels at.

Kang et al. [149] introduce NeRF for hologram generation to avoid the conventional 3D reconstruction process, which is cumbersome, time-consuming, and resolution-limited. The

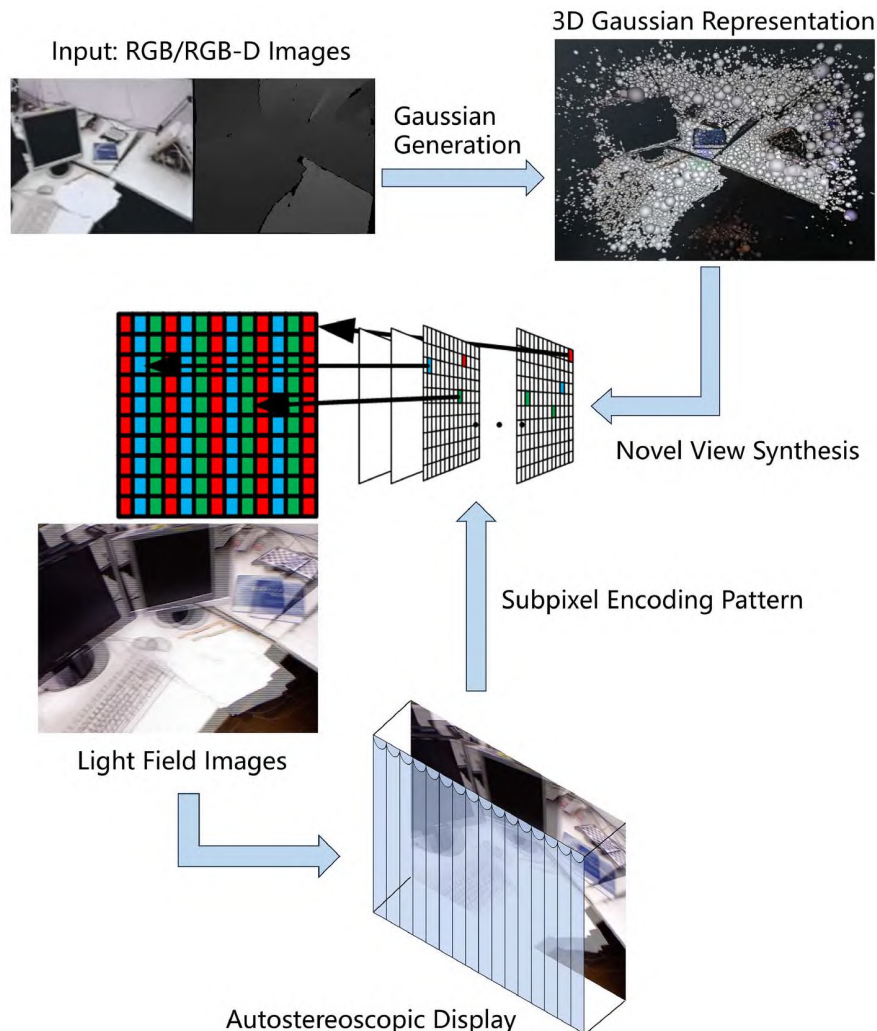


Figure 10 (Color online) The pipeline of radiance field-based light field display.

3D light field predicted by NeRF is used for the new view image rendering and depth inference. Then the RGB images and the corresponding depth images are used for hologram prediction via Tensor Holography [150]. Finally, the 3D images are reproduced and presented in both computer simulation and optical experiment. Choi et al. [151] introduce Gaussian wave splatting (GWS) to achieve an analytical Gaussian-to-Hologram transform supporting occlusions and alpha blending. This CGH algorithm can be implemented efficiently in the Fourier domain, achieving 30× speedup on CUDA kernels.

In stereo display based on light field display or holography, more native and natural novel-view rendering helps the realization of 3D vision, whether through binocular stereo, ray-based methods, or wavefront-based approaches. Current pipelines struggle to independently complete the processes of the acquisition of raw 3D data and the generation of the data in a special format for 3D display. Modern 3D reconstruction provides a natural data process pipeline with image-level or pixel-level data as input and output. The radiance field-based representation acts as a bridge to connect the real space and the displaying space, and the optimization of it is acceptable in terms of time and storage.

Future work can be conducted in the following aspects: (1) Real-time, high-resolution rendering for dynamic content. Research should explore efficient neural representations, sparse Gaussian updates, hardware-aware acceleration, and temporal interpolation to enable high-fidelity rendering (often 4K+ per view) of dynamic scenes. (2) Integration with wavefront and holographic encoding. For holographic displays, future systems could extend to directly predict complex amplitudes or phase-encoded Gaussians, enabling end-to-end optimization from scene representation to SLM driving signals. (3) Calibration-aware and display-integrated training. End-to-end co-design of the 3D representation and the physical display characteristics (such as SLM pixel pitch, diffraction limits, viewing zone) can compensate for optical imperfections. Research should explore display-aware NeRF or 3DGS training that incorporates the forward model of the target holographic or light field hardware.

6 Conclusion

Radiance field-based 3D reconstruction offers a novel paradigm for representing and extracting 3D information from multi-view images. By leveraging learning-based optimization, radiance fields enable the rendering of high-resolution, photorealistic outputs. This paper introduces and systematically compares the original NeRF and 3DGS across multiple dimensions, while also exploring methodologies for extracting geometry from radiance fields. Beyond novel view synthesis and geometry reconstruction, this cutting-

edge technology provides a flexible scene representation framework that can be adapted to various technical fields and application scenarios. Potential directions are discussed with emerging works highlighted and future improvements analysed. Radiance field-based 3D reconstruction is expected to evolve along two main trajectories. The first focuses on developing faster and more efficient reconstruction pipelines for seamless integration into existing application frameworks. The second aims to create more versatile scene representations that support deformation, segmentation, and customized rendering, thereby addressing challenges in interdisciplinary domains such as advanced 3D content generation, processing, and display technologies.

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